DAVID CHO

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EDUCATION

DUKE UNIVERSITY — Durham, NC
PhD in Computer Science – Advisor: Boyuan Chen — Sep 2023 – Present

COLUMBIA UNIVERSITY — New York, NY
Master of Science in Computer Science - Vision, Graphics, Interactions, and Robotics Track — May 2022

COLUMBIA UNIVERSITY — New York, NY
Bachelor of Arts in Physics with a Concentration in Computer Science — May 2020

PUBLICATIONS

- **Dong Heon Cho***, Boyuan Chen. RAFL: Generalizable Sim-to-Real of Soft Robots with Residual Acceleration Field Learning. (IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS 2026).
- Kuang Huang*, **Dong Heon Cho***, Boyuan Chen; *These authors contributed equally to this work. Automated Discovery of Continuous Dynamics from Videos. (npj Artificial Intelligence, 2026)
- Gaurav Jain, Yuanyang Teng, **Dong Heon Cho**, Yunhao Xing, Maryam Aziz, Brian A. Smith. “I Want to Figure Things Out”: Supporting Exploration in Navigation for People with Visual Impairments. (ACM CSCW 2023).
- Peter Yichen Chen, Jinxu Xiang, **Dong Heon Cho**, Yue Chang, G A Pershing, Henrique Teles Maia, Maurizio Chiaramonte, Kevin Carlberg, Eitan Grinspun. CROM: Continuous Reduced-Order Modeling of PDEs Using Implicit Neural Representations. (ICLR 2023, notable-top-25%)

RESEARCH EXPERIENCE

DUKE UNIVERSITY COMPUTER SCIENCE DEPARTMENT — Durham, NC
GENERAL ROBOTICS LAB — Sep 2023 – Present
- Investigated data-driven dynamical systems methods for scientific discovery and generalizable sim-to-real transfer.

COLUMBIA UNIVERSITY COMPUTER SCIENCE DEPARTMENT — New York, NY
COLUMBIA GRAPHICS GROUP — Sep 2021 – Aug 2022
- Developed novel, discretization agnostic, continuous reduced-order modeling (ROM) method to solve PDEs efficiently via implicit neural representations.

COLUMBIA UNIVERSITY COMPUTER SCIENCE DEPARTMENT — New York, NY
COMPUTER ENABLED ABILITIES LAB — Jan 2021 – Aug 2021
- Analyzed visually impaired people (VIP) needs and socio-technical challenges for future Navigation Assistance System (NAS) that facilitate independent exploration.

COLUMBIA UNIVERSITY PHYSICS DEPARTMENT — New York, NY
CONDENSED MATTER LAB — May 2018 – Jul 2018
- Explored semiconducting properties of NbSe2 using scanning tunneling microscopy (STM).

PROFESSIONAL EXPERIENCE

DUKE UNIVERSITY COMPUTER SCIENCE DEPARTMENT, GRAPHICS SOFTWARE ARCHITECTURE - CS345 — Durham, NC
Teaching Assistant — Jan 2024 – May 2024
- Held regular office hours, assisting students in debugging programming assignments in C++.

SAMSUNG RESEARCH AMERICA, THINK TANK TEAM — Mountain View, CA
Research Intern — Jun 2020 – Aug 2020
- Trained three new DOPE-net pose estimation Tensorflow models to detect position and orientation of a specific "object" from RGB images and achieved accuracy within 3cm.
- Produced 300K+ domain-randomized and context aware synthetic data with varying position, lighting, background texture, as well as 3D replicated scene.
- Collaborated with design team to create a custom "object" and determine most accurate initial position to best exhibit pose estimation mechanism for live demo presentation of prototype product (CES 2020).
- Integrated models into ROS Melodic to allow simultaneous detection of multiple “objects” for robotic arm grasping.

TECHNICAL SKILLS

- Python, C, C++, Computational Physics, Java, Swift, TensorFlow, Pytorch, MATLAB, ROS, MySQL, OpenGL